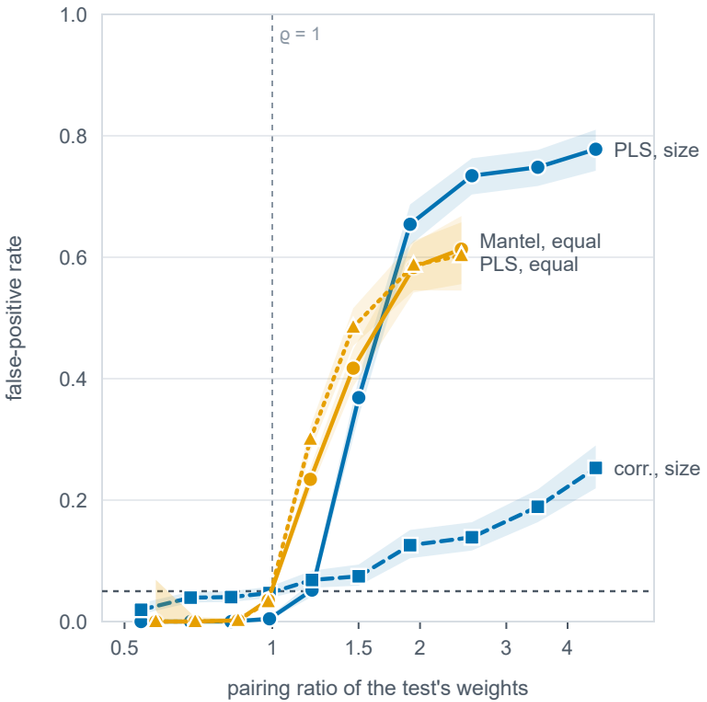
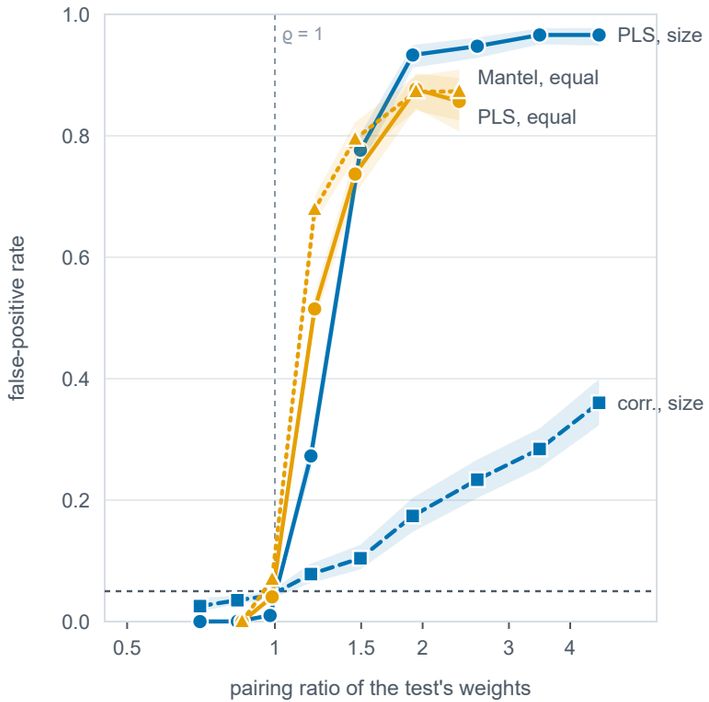
a **M = 20 groups**

each test against the ratio of its own weights



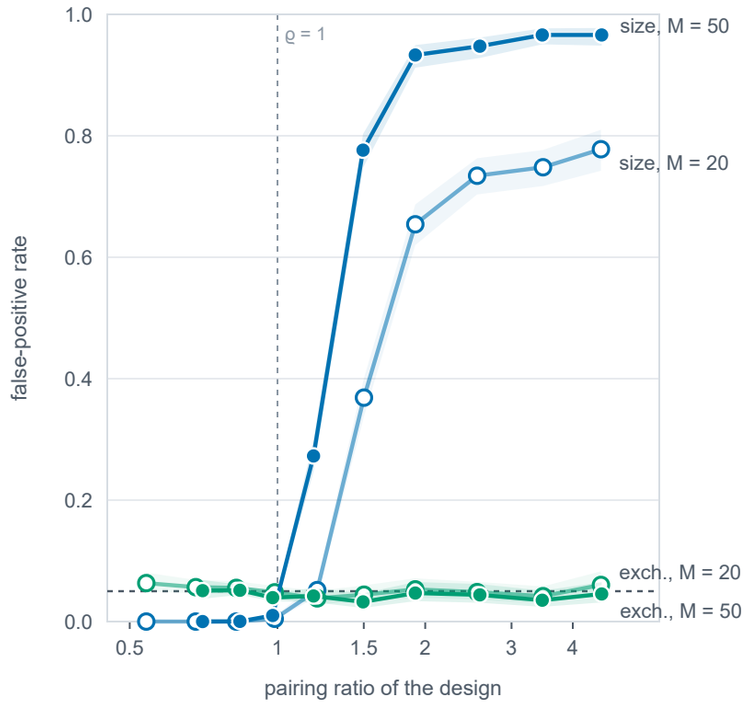
b **M = 50 groups**

each test against the ratio of its own weights



c **Exchangeable weights restore the level**

PLS against the ratio of the design (size weights)



Simulated null data: one group's annotation profile 1.5–4 times as far out as the others, on the largest, the smallest or a random group (24,000 data sets). Dashed line: the nominal rate, 0.05. Bands: 95% intervals per bin of the ratio. Vertical line: ϱ = 1. Exchangeable weights $\propto 1/\sqrt{(\tau^2 + \sigma^2_w / n_m)}$.

Colour: weighting (blue size, orange equal, green exchangeable). Line: test (solid PLS, dashed correlation, dotted Mantel). In c, hollow markers: M = 20; filled: M = 50.